Module Code & Module Title

CU6051NP Artificial Intelligence

Individual Coursework 2

Submission: Milestone 1

Academic Semester: Autumn Semester 2025

Credit: 15 credit semester long module

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Assignment Due Date: 07/01/2026.

Assignment Submission Date: 07/01/2026

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GitHub Link	*https://github.com/Nissan-Shrestha/AI-Coursework.git*

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1. Introduction

A. Artificial Intelligence and Machine Learning

Artificial intelligence is a subdivision of computer science that deals with the creation of systems that can execute those tasks that can be handled by human intelligence and participation. Most of the key fields, including healthcare, finance, transportation, tourism, and others, have already seen the extensive use of AI as of recent years. Machine learning is a branch of AI which helps systems to automatically acquire information and trends and to enhance their work without being explicitly coded (Karimi, Abubakar, Mustafa, & Ahmad, 2024). General categories of machine learning are:

a. Supervised learning

Supervised learning is a form of learning a model based on labelled data, where both the input features and the output labels (labels of the output) are given. The algorithm becomes acquainted with a connection between the inputs and the outputs and applies this acquired connection to make predictions on previously unknown data (Razzaq & Shah, 2025). Classification and regression fall under supervised learning.

b. Unsupervised learning

Unsupervised learning deals with unlabeled data where no output label is available the main goal of unsupervised learning is to discover pattern structures or relationships in the data (Razzaq & Shah, 2025). Clustering and dimensionality reduction are some of the common tasks that fall under unsupervised learning. Clustering is one of the most popular methods of unsupervised learning that tries to cluster similar pieces of data on the basis of their features. Clustering is aimed at having similar data points in the same cluster, more than those in other clusters. The clustering algorithms have found a large number of applications in customer segmentation, image processing, document analysis and in recommendation systems. When it comes to tourism, the concept of clustering can be used to discover groups of tourists sharing similar preferences, behaviour and interests, which allows more personalization and decision-making to be done.

c. Semi-Supervised Learning

In semi-supervised learning, there is a minimal amount of labeled data with a large amount of unlabeled data (Razzaq & Shah, 2025).

d. Reinforcement learning

Reinforcement learning is about learning by means of interacting with the environment. Certain actions are performed which result in either receiving rewards or penalties, with focus of maximizing rewards over time (Razzaq & Shah, 2025).

B. Problem Domain: Tourist Clustering

There are various preferences among tourists in terms of choice of places of destination. Some types of tourism spots like museums and art galleries may attract some tourists, whereas others will be attracted by beaches, resorts, or even nightlife. The fact that all tourists might be treated as one group may lead to unsuccessful recommendations and marketing policies. A tourist interested in cultural destinations is highly likely to be interested in advertisements or packages for cultural spots rather than others. Hence, the tourism industry presents a challenge in determining different segments of tourists in terms of their preferences.

Tourist clustering attempts to solve this issue through clustering the travelers based on their preferences in destinations. These clusters will prove useful in allowing travel agencies, tourism boards and recommendation platforms to draft specific services, customized travel packages and better destination planning strategies.

C. Dataset Overview

The link for the dataset can be found below:

http://archive.ics.uci.edu/dataset/484/travel+reviews

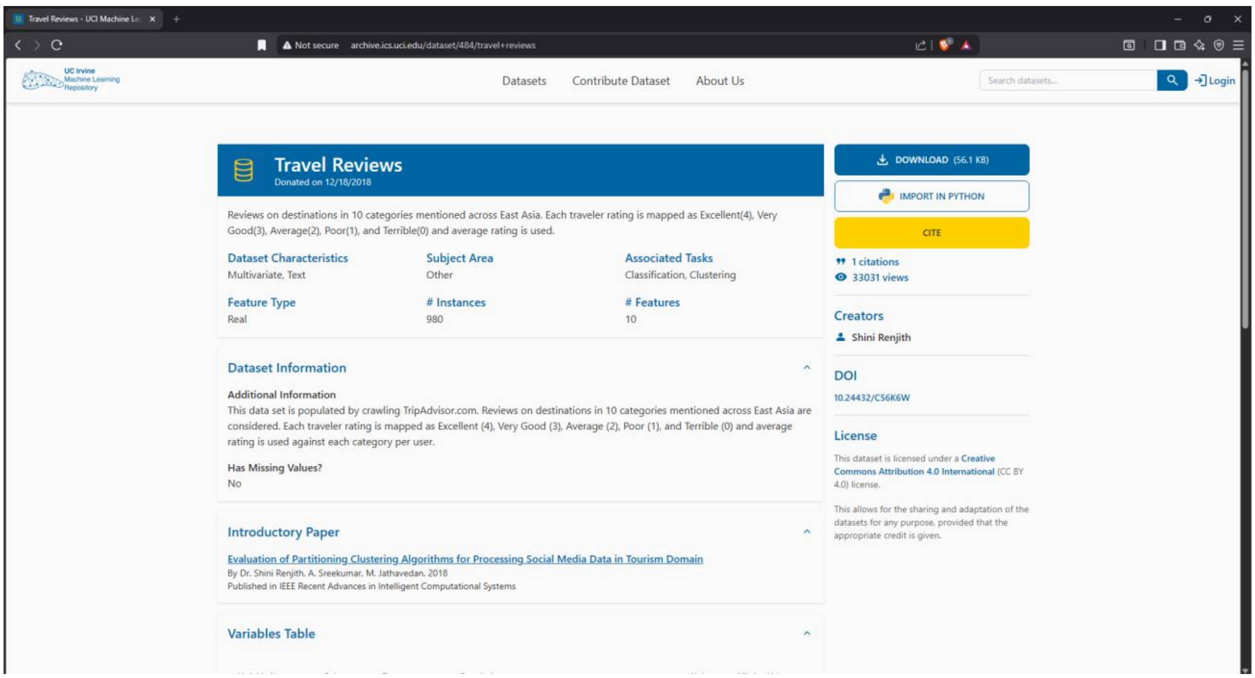

Figure 1 Dataset Used

The dataset used in this coursework is a TripAdvisor based dataset which consists of reviews from tourists that visited East Asian destinations. It contains 980 cases, a case representing data from a single traveler. Every tourist is categorized in terms of 10 numeric characteristics, which corresponds to the average ratings that have been given by the respective traveler about various types of tourist destinations, which include art galleries, dance clubs, juice bars, restaurants, museums, resorts, parks/picnic spots, beaches, theaters and religious institutions. Each rating is mapped as Excellent (4), Very Good (3), Average (2), Poor (1), and Terrible (0).

This dataset does not entail any known class labels therefore an unsupervised learning strategy is suitable. This coursework suggests a solution to segment tourists based on their preferences in clusters, which is based on machine learning methods. Evaluating tendencies in tourist review data, the suggested methodology will help to prove how clustering will assist in real-life tourism applications. This process will result in multiple tourist segments that represent different preference patterns.

Table 1 Data Dictionary

Variable Name	Role	Type	Description
User ID	ID	Categorical	User id for the tourist
Category 1	Feature	Continuous	Average user feedback on art galleries
Category 2	Feature	Continuous	Average user feedback on dance clubs
Category 3	Feature	Continuous	Average user feedback on juice bars
Category 4	Feature	Continuous	Average user feedback on restaurants
Category 5	Feature	Continuous	Average user feedback on museums
Category 6	Feature	Continuous	Average user feedback on resorts
Category 7	Feature	Continuous	Average user feedback on parks/picnic spots
Category 8	Feature	Continuous	Average user feedback on beaches
Category 9	Feature	Continuous	Average user feedback on theaters
Category 10	Feature	Continuous	Average user feedback on religious institutions

2. Background

A. Research done on the problem domain

Multiple studies have already explored the use of machine learning techniques to understand tourist behavior and preferences. As online travel websites become more and more equipped with user-generated content, scholars have applied clustering techniques to define useful segments of tourists.

(Renjith, Sreekumar, & Jathavedan, 2018) conducted the evaluation of partitioning clustering algorithms on processing the social media data within the tourism domain based on the reviews on TripAdvisor. They carried out a study that aimed at grouping tourists according to preference ratings of various destination categories. The authors have established that the clustering methods applied especially the K-Means are useful in identifying specific groups of tourists without the need to have the labeled data. The findings indicated that coherent clusters are likely to develop among tourists who share similar interests, thus unsupervised learning is appropriate when it comes to tourism analytics. The outcomes also highlighted that clustering may help tourism stakeholders to know the pattern of tourist behavior and enhance data-driven decision-making. This study gives a first-hand base to the current study, since it confirms the application of clustering methods on TripAdvisor-based tourism data sets.

(Dolnicar, 2002) has summarised data-based techniques in market segmentation within the tourism industry and highlighted the shortcomings of the traditional manual and demographic based market segmentation techniques. This research had the argument that the traditional forms of segmentation do not usually suit the intricate and varied preferences of the contemporary travellers. It was emphasized that the clustering methods allow making a more precise and objective identification of the segments of tourists analysing their behavioural and preference data instead of basing on the already established assumptions. The study found that the data-driven clustering solutions have a better flexibility and reliability in tourist profiling which confirms the implementation of machine learning to ensure tourism segmentation. This article reinforces theoretical rationale of using clustering methods in research on tourism.

(Xiang & Gretzel, 2010) also analysed how social media is utilized in online travel information search and emphasized the increasing relevance of user-generated information in the context of studying tourist behaviour. They conducted a study of the dependence of the online ratings, reviews, and shared experiences by tourists in making their travel decisions. The authors have shown that social media sites produce big amounts of useful data capable of being scrutinized to get information about the preferences and the level of satisfaction of tourists. The study also implied that the study of online reviews can assist in personalisation, narrow-market marketing and higher levels of tourism service delivery. Such results support the importance of using clustering methods on tourism data obtained with the help of online services, including the TripAdvisor, to draw useful patterns in user-generated information.

B. Review and Analysis of Existing Work

Although current studies prove the usefulness of clustering methods in tourism analytics, some drawbacks can be traced through the studies carried out before. The majority of studies, such as the one of (Renjith, Sreekumar, & Jathavedan, 2018) mainly concentrate on the algorithms of partition-based clustering like K-Means because they are computationally efficient. Nevertheless, K-Means needs to be specified in advance how many clusters are to be used and is sensitive to the initial choice of centroids which can influence the consistency of results when used on complex tourism data.

Conceptual tourism work like (Dolnicar, 2002) has suggested the significance of meaningful segmentation of tourists but lacks algorithmic answers to large-scale, preference-based data analysis. Although the research is a very convincing argument in favor of data-based segmentation, it also brings up the fact that clusters are difficult to explain in a manner that is practically applicable to tourism players.

In addition, (Xiang & Gretzel, 2010) also discuss the increased role of user-generated content in terms of tourism decision making without examining the methods of analyzing such information. This indicates the lack of connection between the presence of rich tourism data and the use of advanced machine learning techniques to process it.

On the basis of this analysis, it is evident that there is a requirement of an approach that is not only applicable in efficient clustering algorithm but enhances interpretability and validation of tourist segments. Thus, this coursework suggests adopting the concept of employing various clustering algorithms, such as K-Means and Hierarchical Clustering, to overcome the drawbacks of the current research and offer more strong and understandable tourist segmentation.

3. Proposed Solution

The aim of the solution suggested is to find meaningful tourist segments, according to their preferences regarding the type of destinations. As no pre-defined labels are present in the dataset and the analysis is aimed at the average rating of the users in several categories, an unsupervised variant of learning is chosen.

The solution proposed has a systematic clustering pipeline. To begin with, the TripAdvisor based tourism data is examined to comprehend its structure and characteristics. The dataset is arranged in such a way that each row corresponds to one tourist whereas each column corresponds to the average rating that the tourist gives to a particular category of destinations. Since the dataset has numerical values on a homogenous scale and no blank values, it can be subjected to distance-based clustering.

Clustering algorithms are then used to place tourists having similar preference patterns together. In order to enhance robustness and interpretability, more than one clustering algorithm has been applied. Internal validation methodologies, i.e. the elbow method and silhouette analysis are used to measure the number of clusters and domain knowledge that is associated with tourism segmentation.

Lastly, the resulting clusters are understood through examining the most common categories of destinations in each cluster. These groups would also be representative of different segments of tourists, like culture perspective, leisure perspective, or entertainment perspective tourists. What is being proposed is focused on interpretability and relevance to the field of life, as opposed to predictive accuracy.

A. AI algorithms used

a. K-Means Clustering

K-Means is a partition based unsupervised learning algorithm that clusters data points in a predetermined number of clusters depending on their similarity (Rahaman, 2024). The algorithm operates by setting a centroid of each of the clusters and finds each piece of data to the nearest centroid by an iterative process of assigning each data point to the nearest centroid based on a distance parameter, usually the Euclidean distance. Centroids will then be updated as the average of the data points allotted to each cluster.

K-Means is selected to be used in present research because of its simplicity, efficiency, and the ability to work with numerical data like tourist rating data. Nevertheless, since K-Means requires the number of clusters to be predetermined, internal validation procedures are applied to estimate a suitable value of K.

b. Hierarchical Clustering

The other unsupervised learning method is hierarchical clustering which constructs the hierarchy of clusters. In the current work, an agglomerative hierarchical design is taken into consideration, whereby each tourist is assumed to create a cluster first, and clusters are later merged depending on the similarity. Hierarchical clustering generates a dendrogram that visually depicts the process of merging items and enables one to understand relationships of clusters better (Noble). This algorithm is employed together with K-Means to assist in cluster validation and enhance interpretability. The consistency and the stability of tourist segments can be tested by comparing the results of both algorithms.

B. Pseudocode

Input:

Dataset D containing n tourists

Each tourist is described by m destination rating features

Output:

Grouped tourist segments based on preference similarities

Begin

// Data Understanding and Preparation

Remove non-informative attributes (User ID)

Select destination rating features for clustering

Normalize all rating features to ensure equal contribution

// Determine Optimal Number of Clusters for K-Means

Define a range of cluster numbers K = 2 to K_max

For each value of K in the defined range **do**:

Randomly **initialize** K cluster centroids

Repeat:

For each tourist record i in dataset D **do:**

Compute distance between record i and each centroid

Assign record i to the nearest centroid

End For

For each cluster j do:

Recalculate centroid j as the mean of all records assigned to it

End For

Until centroids converge or no changes occur

Calculate internal validation metrics

- Within-Cluster Sum of Squares (WCSS)

- Silhouette score

End For

Analyze validation results to select the optimal value of K

// Final K-Means Clustering

Apply K-Means algorithm using the selected optimal K

Assign each tourist to a final cluster

// Hierarchical Agglomerative Clustering

Initialize each tourist as an individual cluster

While more than one cluster exists do:

Compute distances between all existing clusters

Identify the two most similar clusters

Merge the two closest clusters into a single cluster

End While

Generate dendrogram to represent hierarchical cluster structure

// Cluster Interpretation

Determine final number of clusters by cutting the dendrogram

Analyze average ratings within each cluster

Label clusters based on dominant tourist preferences

End

C. Diagrammatical representations of the solution

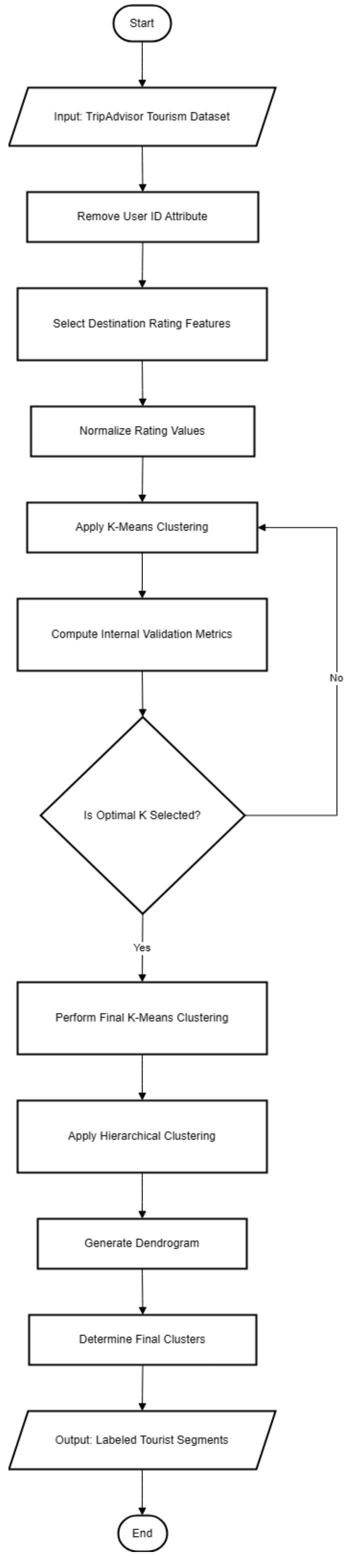

Figure 2 Flowchart for the solution

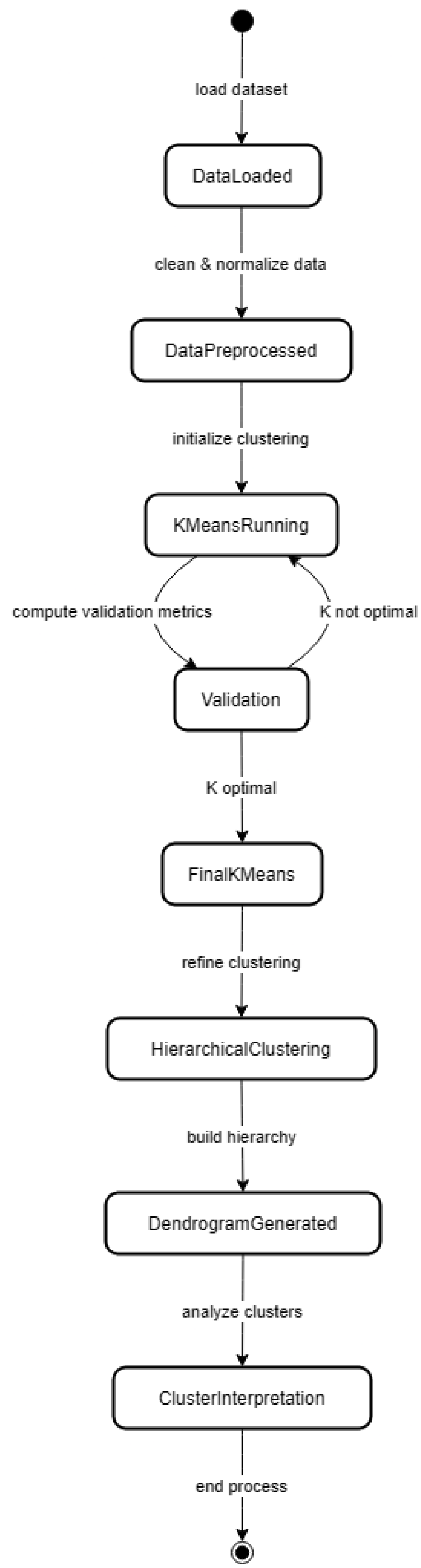

Figure 3 State Transition Diagram for the solution

4. Development Process

```
[114]: import pandas as pd
import numpy as np
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
import matplotlib.pyplot as plt
```

Figure 4 Importing required libraries

The following Python libraries and Libraries were imported to build the application:

1. Pandas
    - Used for reading, cleaning and manipulating tabular data.
2. NumPy
    - Provides numerical operations and array handling.
3. Matplotlib
    - Python library used to create 2D plots and visualizations.
4. Seaborn
    - A high-level visualization library based on Matplotlib.
5. Scikit_learn(sklearn)
    - Machine learning library for data preprocessing and clustering.
6. Scipy (hierarchy module)
    - Provides hierarchical clustering functions.

The development starts with the analysis of the appropriate Python libraries that are to be imported in order to manipulate, visualize, and train the data. The tripadvisorreview.csv data is loaded into a Dataset, which is a pandas DataFrame with 980 entries of tourists. Every tourist is graded in 10 categories of attraction, which were first identified as generic (Category 1 - Category 10). These groups get directly renamed into more qualitative names: Art Galleries, Dance Clubs, Juice Bars, Restaurants, Museums, Resorts, Parks, Beaches, Theatres, and Religious Institutions. The preliminary data analysis demonstrates that there are no values missing and the data structure is regular having all the rating columns with the same type of float data except the UserID identifier.

```
]: data = pd.read_csv("tripadvisor_review.csv")

   data.head()
```

]:

	User ID	Category 1	Category 2	Category 3	Category 4	Category 5	Category 6	Category 7	Category 8	Category 9	Category 10
0	User 1	0.93	1.8	2.29	0.62	0.80	2.42	3.19	2.79	1.82	2.42
1	User 2	1.02	2.2	2.66	0.64	1.42	3.18	3.21	2.63	1.86	2.32
2	User 3	1.22	0.8	0.54	0.53	0.24	1.54	3.18	2.80	1.31	2.50
3	User 4	0.45	1.8	0.29	0.57	0.46	1.52	3.18	2.96	1.57	2.86
4	User 5	0.51	1.2	1.18	0.57	1.54	2.02	3.18	2.78	1.18	2.54

Figure 5 Loading the dataset

```
data.columns = [
    'User_ID', 'Art_Galleries', 'Dance_Clubs', 'Juice_Bars', 'Restaurants',
    'Museums', 'Resorts', 'Parks', 'Beaches', 'Theaters', 'Religious_Institutions'
]

# Display first 5 rows to check
data.head()
```

	User_ID	Art_Galleries	Dance_Clubs	Juice_Bars	Restaurants	Museums	Resorts	Parks	Beaches	Theaters	Religious_Institutions
0	User 1	0.93	1.8	2.29	0.62	0.80	2.42	3.19	2.79	1.82	2.42
1	User 2	1.02	2.2	2.66	0.64	1.42	3.18	3.21	2.63	1.86	2.32
2	User 3	1.22	0.8	0.54	0.53	0.24	1.54	3.18	2.80	1.31	2.50
3	User 4	0.45	1.8	0.29	0.57	0.46	1.52	3.18	2.96	1.57	2.86
4	User 5	0.51	1.2	1.18	0.57	1.54	2.02	3.18	2.78	1.18	2.54

Figure 6 Changing Column Names

```python
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 980 entries, 0 to 979
Data columns (total 11 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   User_ID                980 non-null    object
 1   Art_Galleries          980 non-null    float64
 2   Dance_Clubs            980 non-null    float64
 3   Juice_Bars             980 non-null    float64
 4   Restaurants            980 non-null    float64
 5   Museums                980 non-null    float64
 6   Resorts                980 non-null    float64
 7   Parks                  980 non-null    float64
 8   Beaches                980 non-null    float64
 9   Theaters               980 non-null    float64
 10  Religious_Institutions 980 non-null    float64
dtypes: float64(10), object(1)
memory usage: 84.3+ KB
```

```python
data.isnull().sum()
```

```
User_ID                   0
Art_Galleries             0
Dance_Clubs               0
Juice_Bars                0
Restaurants               0
Museums                   0
Resorts                   0
Parks                     0
Beaches                   0
Theaters                  0
Religious_Institutions    0
dtype: int64
```

Figure 7 Exploring the dataset and checking for null values

Prior to the operation of clustering algorithms, the data is pre-processed in a necessary way. The column of UserID is eliminated as only numerical features are required by clustering algorithms. More to the point, all rating columns are normalized by MinMaxScaler that converts each feature to the range of 0-1. This scaling is essential as clustering methods such as K-Means are susceptible to the size of features, and without scaling these categories, with naturally higher values of ratings, may have an inappropriate impact on the cluster formation process. The resulting scaled dataset is such that each category of attractions makes equal contribution to distance process which is the basis of the two clustering methods.

[119]:
```
data_clean = data.drop(columns=['User_ID'])
data_clean.head()
```

[119]:

	Art_Galleries	Dance_Clubs	Juice_Bars	Restaurants	Museums	Resorts	Parks	Beaches	Theaters	Religious_Institutions
0	0.93	1.8	2.29	0.62	0.80	2.42	3.19	2.79	1.82	2.42
1	1.02	2.2	2.66	0.64	1.42	3.18	3.21	2.63	1.86	2.32
2	1.22	0.8	0.54	0.53	0.24	1.54	3.18	2.80	1.31	2.50
3	0.45	1.8	0.29	0.57	0.46	1.52	3.18	2.96	1.57	2.86
4	0.51	1.2	1.18	0.57	1.54	2.02	3.18	2.78	1.18	2.54

Figure 8 Dropping User Id column

```
scaler = MinMaxScaler()

scaled_data = scaler.fit_transform(data_clean)

scaled_df = pd.DataFrame(scaled_data, columns=data_clean.columns)
scaled_df.head()
```

	Art_Galleries	Dance_Clubs	Juice_Bars	Restaurants	Museums	Resorts	Parks	Beaches	Theaters	Religious_Institutions
0	0.204861	0.494505	0.618911	0.142857	0.228395	0.629834	0.6	0.381443	0.444444	0.184211
1	0.236111	0.604396	0.724928	0.148936	0.419753	0.839779	1.0	0.216495	0.460905	0.118421
2	0.305556	0.219780	0.117479	0.115502	0.055556	0.386740	0.4	0.391753	0.234568	0.236842
3	0.038194	0.494505	0.045845	0.127660	0.123457	0.381215	0.4	0.556701	0.341564	0.473684
4	0.059028	0.329670	0.300860	0.127660	0.456790	0.519337	0.4	0.371134	0.181070	0.263158

Figure 9 Normalizing the ratings

The K-Means clustering consists of planning the best number of clusters based on the Elbow Method. In this method, K-Means is run with different values of k (between 2 and 10) and a plot of Within-Cluster Sum of Squares (WCSS) against k is plotted. The point at which the rate of decrease in WCSS is no longer favoured is the elbow point indicating the best k value. The analysis is then carried out with k= 6 that seems to be acceptable according to the elbow plot. The silhouette score of this configuration is approximately 0.16, which means that the separation of clusters is rather weak but yet allows performing an exploratory analysis. K-Means algorithm is then used in practice, with 6 clusters and a deterministic random state, so that every tourist falls into one of six different categories.

```
•[135]: wcss = []

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(scaled_data)
    wcss.append(kmeans.inertia_)

plt.plot(range(2, 11), wcss, marker='o')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method to Determine Optimal K")
plt.show()
```

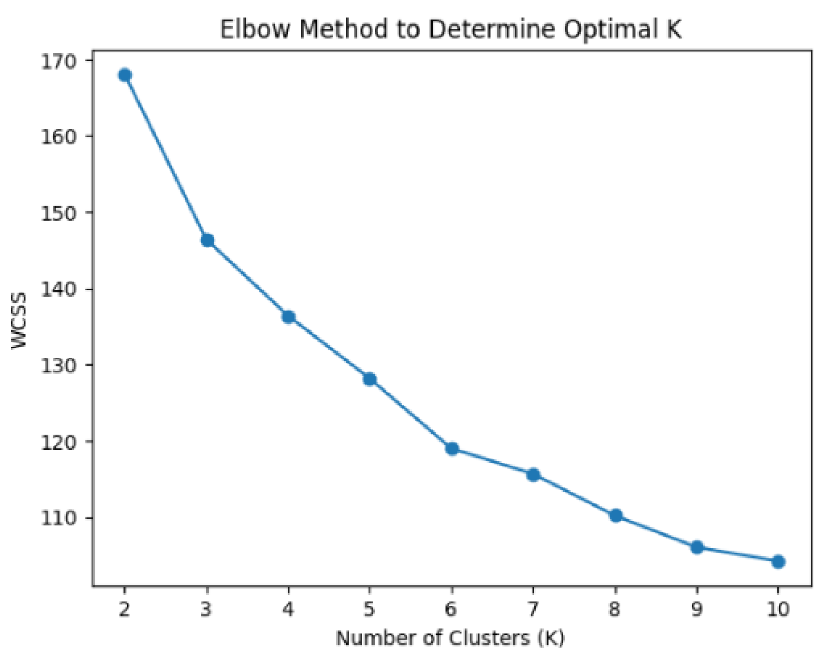

Figure 10 Determining optimal k

```
kmeans_temp = KMeans(n_clusters=6, random_state=42)
temp_labels = kmeans_temp.fit_predict(scaled_data)

silhouette_avg = silhouette_score(scaled_data, temp_labels)
silhouette_avg
```

```
0.1618113212405641
```

Figure 11 Calculating Silhouette Score

```
kmeans = KMeans(n_clusters=6, random_state=42)

cluster_labels = kmeans.fit_predict(scaled_data)

data_clean['Cluster'] = cluster_labels

data_clean.head()
```

	Art_Galleries	Dance_Clubs	Juice_Bars	Restaurants	Museums	Resorts	Parks	Beaches	Theaters	Religious_Institutions	Cluster
0	0.93	1.8	2.29	0.62	0.80	2.42	3.19	2.79	1.82	2.42	1
1	1.02	2.2	2.66	0.64	1.42	3.18	3.21	2.63	1.86	2.32	1
2	1.22	0.8	0.54	0.53	0.24	1.54	3.18	2.80	1.31	2.50	3
3	0.45	1.8	0.29	0.57	0.46	1.52	3.18	2.96	1.57	2.86	3
4	0.51	1.2	1.18	0.57	1.54	2.02	3.18	2.78	1.18	2.54	4

Figure 12 Applying K-Means Algorithm

After the cluster assignment, the analysis investigates the features of each cluster. The sizes of the clusters show skewed distribution where Cluster 3 has the highest number of tourists (217) and Cluster 5 the least number of tourists (116). To know what is unique to each cluster, the mean rating of each category of attractions is done in each cluster. These averages demonstrate evident favourable tendencies. As an example, the cluster with the highest ratings in terms of Religious Institutions and Museums can be seen to indicate the cultural and religious tourists. The other cluster is highly engaged in almost all the categories, which points to broad-interest tourists. According to these patterns, the clusters obtain interpretative titles which include Nature and Beach-Oriented Tourist, Entertainment and Leisure Seeker, Religious and Cultural Tourist, Balanced Sightseeing Tourist, High-Engagement Multi- Interests Tourist and Relaxed Leisure Tourist.

```
data_clean['Cluster'].value_counts()
```

```
Cluster
3    217
2    187
1    176
4    147
0    137
5    116
Name: count, dtype: int64
```

Figure 13 Number of tourists in each cluster

```
kmeans_means = data_clean.groupby('Cluster').mean()
kmeans_means
```

Cluster	Art_Galleries	Dance_Clubs	Juice_Bars	Restaurants	Museums	Resorts	Parks	Beaches	Theaters	Religious_Institutions
0	0.862044	1.208467	0.579197	0.510803	0.970365	1.953723	3.179781	2.996642	1.992117	2.844161
1	0.872330	1.459773	2.259773	0.595568	1.131250	2.170341	3.193239	2.791477	1.585057	2.462500
2	0.933797	1.332193	0.418021	0.386845	0.755508	1.434011	3.170963	2.838770	1.463957	3.256043
3	0.839908	1.329217	0.390968	0.593687	0.673088	1.582120	3.179631	2.812396	1.542304	2.714700
4	0.970272	1.527075	1.174150	0.649796	1.356599	2.421633	3.182313	2.806190	1.519456	2.564898
5	0.898190	1.215862	1.554828	0.434138	0.880517	1.628793	3.180431	2.783362	1.330690	2.975690

Figure 14 Calculating mean rating for each category within each cluster

[126]:
```
kmeans_labels = {
    0: "Nature & Beach-Oriented Tourists",
    1: "Entertainment & Leisure Seekers",
    2: "Religious & Cultural Tourists",
    3: "Balanced Sightseeing Tourists",
    4: "High-Engagement Multi-Interest Tourists",
    5: "Relaxed Leisure Tourists"
}

data_clean['Cluster_Label'] = data_clean['Cluster'].map(kmeans_labels)
```

[127]:
```
data_clean[['Cluster', 'Cluster_Label']].head(10)
```

[127]:

	Cluster	Cluster_Label
0	1	Entertainment & Leisure Seekers
1	1	Entertainment & Leisure Seekers
2	3	Balanced Sightseeing Tourists
3	3	Balanced Sightseeing Tourists
4	4	High-Engagement Multi-Interest Tourists
5	2	Religious & Cultural Tourists
6	2	Religious & Cultural Tourists
7	3	Balanced Sightseeing Tourists
8	4	High-Engagement Multi-Interest Tourists
9	2	Religious & Cultural Tourists

Figure 15 Labelling the clusters

The dataset was then subjected to hierarchical clustering in order to investigate the inherent structure of the data without necessarily having pre-specified clusters. Before the clustering, all the numerical features were first made to follow Min-Max scaling so that every attribute would play the same role in the distance calculations.

The hierarchical clustering was used based on the Ward method of linkage and was called agglomerative. This method treats the individual observational clusters as clusters and repeatedly combines the nearest clusters in terms of minimizing the increment of within-cluster variance. The reason why Ward method was chosen was due to its ability to give small and interpretable clusters which is useful in continuous numerical data.

The scaled data set was used to calculate the linkage matrix and a dendrogram was drawn to give a visual representation of the hierarchical organization of the data. The dendrogram helped in understanding the relationship between the clusters and helped in deciding the number of clusters to have as it was found that the vertical distances between the merged clusters rose significantly at a given level.

```
linked = linkage(scaled_data, method='ward')
plt.figure(figsize=(12, 6))
dendrogram(linked, truncate_mode='lastp', p=12, leaf_rotation=90., leaf_font_size=10.)
plt.xlabel("Cluster Index")
plt.ylabel("Euclidean Distance")
plt.title("Hierarchical Clustering Dendrogram (Truncated)")
plt.show()
```

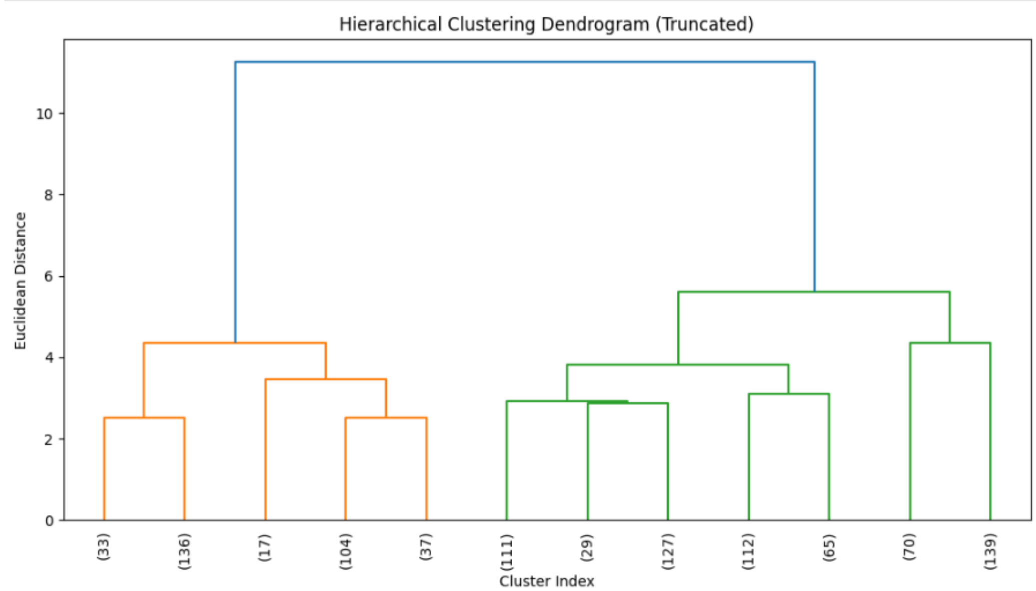

Figure 16 Constructing Dendrogram

Despite visual inspection of the dendrogram indicating that possibly a small number of clusters could be used to extract the highest level of separation in the data, hierarchical clustering solution was pruned to generate six clusters. This was selected to be consistent with the number of clusters determined with the elbow method during the K-Means analysis and allow direct comparison of the two cluster methods.

A maximum cluster criterion was used to assign cluster membership and hierarchical cluster labels were imposed on the original (unscaled) dataset to use in an interpretative analysis. In order to know the attributes of each cluster, mean values of the features related to tourism were taken in all the hierarchical clusters. These cluster profiles were then employed to give descriptive names which capture the prevailing tourist preferences, including activities that are nature oriented, culture and religious interests and activities that are leisure oriented.

The hierarchical clustering solution formed six clusters of different sizes, which indicated the flexibility of the algorithm to form both common and niche segments of tourists. Cluster sizes were between 70 to 267 observations and one larger cluster made of a widely shared tourist profile and smaller clusters of more specific preference patterns. The lack of the very small or fragmented clusters indicates that the hierarchical solution is stable and is interpretable.

The hierarchical clustering outcomes showed larger and more diverse segments of tourists than those of K-Means, which is a feature of the algorithm as it is based on the structural connection and not on explicit optimization of the cluster-clusters distance. This is supported further by a lower silhouette score compared to K-Means meaning that there is more overlap in clusters. However, hierarchical clustering offered a good background information of the nest of preferences of tourists, and it served as a complement to the K-Means analysis to identify other groupings in the data.

```
[142]: hierarchical_labels = fcluster(linked, 6, criterion='maxclust')
hierarchical_df = data_clean.copy()
hierarchical_df = hierarchical_df.drop(columns=['Cluster', 'Cluster_Label'], errors='ignore')

hierarchical_df['Hierarchical_Cluster'] = hierarchical_labels
hierarchical_df.head()
```

[142]:

	Art_Galleries	Dance_Clubs	Juice_Bars	Restaurants	Museums	Resorts	Parks	Beaches	Theaters	Religious_Institutions	Hierarchical_Cluster
0	0.93	1.8	2.29	0.62	0.80	2.42	3.19	2.79	1.82	2.42	1
1	1.02	2.2	2.66	0.64	1.42	3.18	3.21	2.63	1.86	2.32	1
2	1.22	0.8	0.54	0.53	0.24	1.54	3.18	2.80	1.31	2.50	4
3	0.45	1.8	0.29	0.57	0.46	1.52	3.18	2.96	1.57	2.86	4
4	0.51	1.2	1.18	0.57	1.54	2.02	3.18	2.78	1.18	2.54	2

Figure 17 Assigning Cluster Membership and imposing labels

```
]: hier_silhouette_avg = silhouette_score(scaled_data, hierarchical_labels)
hier_silhouette_avg
```

```
]: 0.11755988663962905
```

Figure 18 Calculating Silhouette Score

```
hierarchical_means = hierarchical_df.groupby('Hierarchical_Cluster').mean()
hierarchical_means
```

Hierarchical_Cluster	Art_Galleries	Dance_Clubs	Juice_Bars	Restaurants	Museums	Resorts	Parks	Beaches	Theaters	Religious_Institutions
1	0.843373	1.484734	2.248876	0.569882	1.150178	2.151124	3.192485	2.768698	1.534438	2.470178
2	0.909241	1.466329	1.340696	0.734114	1.265823	2.301709	3.183797	2.851646	1.576772	2.568861
3	0.825618	1.407041	0.583633	0.492210	0.915581	1.837191	3.179139	2.905318	1.855243	2.882060
4	0.972938	1.175141	0.463107	0.544011	0.658644	1.577966	3.179322	2.796836	1.457740	2.733616
5	0.904286	1.218857	1.576571	0.419571	0.787429	1.545714	3.180286	2.750286	1.209000	3.036571
6	0.958201	1.251511	0.381223	0.377482	0.794245	1.444604	3.169496	2.853309	1.378417	3.266043

Figure 19 Calculating Mean

```
hierarchical_df['Hierarchical_Cluster'].value_counts()
```

```
Hierarchical_Cluster
3    267
4    177
1    169
2    158
6    139
5     70
Name: count, dtype: int64
```

Figure 21 Number of tourists in each cluster

```
hier ... (10)
```

	H	
0		
1		
2		
3		
4	2	Social & Nightlife Seekers
5	6	Casual Leisure & Dining Tourists
6	6	Casual Leisure & Dining Tourists
7	6	Casual Leisure & Dining Tourists
8	2	Social & Nightlife Seekers
9	3	Spiritual & Nature-Focused Tourists

Figure 20 Labelling the clusters

5. Comparison of Clustering Algorithms

To measure and compare the clustering performance, K- Means and hierarchical clustering were done on the same scaled data with six clusters. This guaranteed methodological consistency and was able to directly compare the two ways.

Depending on the clusters that K-Means clustering generated, relatively compact and well-separated clusters were formed, which is indicated by the fact that its silhouette score was higher. This is not surprising given the fact that K-Means is specifically optimized to maximize within-cluster variance and thus it prefers closer groupings and more distinct cluster boundaries. Consequently, the K-Means clusters were more homogenous and even based, which is why they can be used when the fine-grained segmentation of the tourist preferences is required.

On the contrary, the hierarchical clustering gave larger and more heterogeneous clusters. Hierarchical clustering, in contrast to K-Means, is not directly maximizing the separation of clusters but, rather, clusters are formed on the basis of structural relationships among observations. Such an attribute resulted in larger overlap of clusters as the silhouette score was smaller. Nevertheless, hierarchical clustering was good at exposing larger categories of tourists and describing embedded relationships in the data that were not as distinct in the K-Means solution.

[149]:
```
silhouette_comparison = pd.DataFrame({
    "Clustering Method": ["K-Means", "Hierarchical"],
    "Silhouette Score": [silhouette_avg, hier_silhouette_avg]
})

silhouette_comparison
```

[149]:

	Clustering Method	Silhouette Score
0	K-Means	0.161811
1	Hierarchical	0.117560

Figure 22 Comparing Silhouette Score

The association between the two clustering outcomes was also explored with the help of a cross-tabulation and a heatmap. These illustrations showed that although there is some correspondence among the clusters generated by the two algorithms, there still exist significant disparities in the way tourists are clustered. It implies that the techniques put an emphasis on dissimilarities and differences of the data, K-Means prioritizes compactness and separation, and hierarchical clustering prioritizes similarity in the structure.

```
comparison_table = pd.crosstab(
    data_clean['Cluster'],
    hierarchical_df['Hierarchical_Cluster']
)
comparison_table
```

Hierarchical_Cluster	1	2	3	4	5	6
Cluster						
0	0	9	121	5	0	2
1	152	23	0	1	0	0
2	0	0	35	13	4	135
3	0	12	63	140	0	2
4	13	99	25	9	1	0
5	4	15	23	9	65	0

Figure 23 Cross Tabulation

```python
plt.figure(figsize=(8,6))
sns.heatmap(comparison_table, annot=True, fmt="d", cmap="YlGnBu")
plt.title("Comparison: K-Means vs Hierarchical Clusters")
plt.xlabel("Hierarchical Cluster")
plt.ylabel("K-Means Cluster")
plt.show()
```

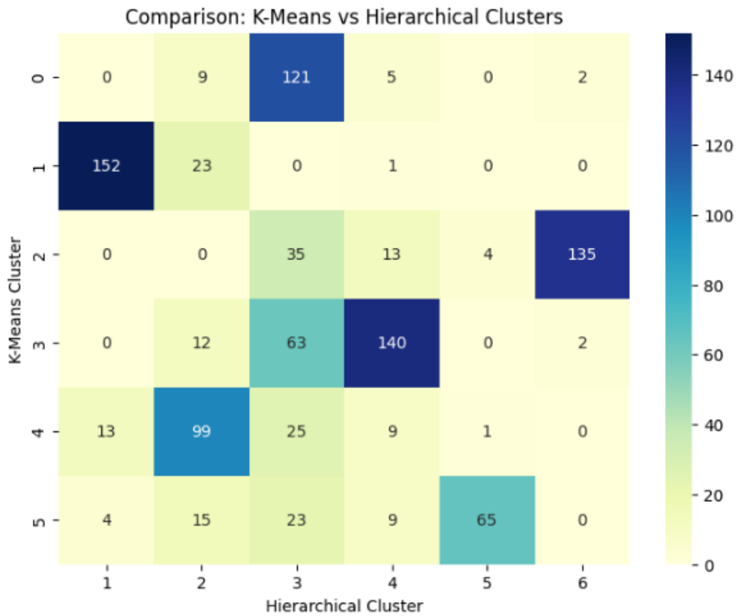

Figure 24 Heat Map

Generally, K-Means was more suitable in producing distinct and understandable clusters whereas the hierarchical clustering had supplementary information on the structure of tourist preferences. The two approaches collectively provide a better insight into the dataset assigned compared to each one of them individually.

6. Cluster Interpretation

A. K-Means Clustering

The obtained silhouette score of 0.162 shows that the clusters are not completely separated, but there is an intermediate grouping structure in the data, which shows the patterns of similarity in tourist activity preferences among the different locations. The clusters differ in size with Cluster 3 having the most number of observations of 217, and Cluster 5 having the least numbers of 116. This difference implies that some forms of tourist preference are more prevalent whereas others are more of a niche or specialized behaviours.

A closer look at the cluster centroids will give more information about the character of each group. Cluster 0, denoted as Nature and Beach-oriented Tourists, is defined as high scores in Parks and Beaches, and moderate scores of activities in Art Galleries, dance clubs and museums. It means that the tourists in this segment are mainly after outdoor and natural attractions and moderately involved in cultural activities as well. Cluster 1, which is called Entertainment and Leisure Seekers, indicates a high level in Juice Bars, Dance Clubs, and Resorts and moderate level in Restaurant activity, and this reflects tourists who like to night life, leisure, and entertainment. Cluster 2 identified as the Religious and Cultural Tourists has a greater value in Religious Institution at a lower value in both entertainment and eating facilities, indicating a cultural or spiritual over the recreational. Cluster 3, the Balanced Sightseeing Tourists, exhibits moderate values on almost all features and this represents the tourists who have a combination of all the three in terms of sightseeing, cultural experiences and outdoor activity, but not any one activity overpowering the rest. Cluster 4, High-Engagement Multi-Interest Tourists, is characterized by the high values of Museums, Resorts and Dance Clubs, which are the tourists who pursue the combination of cultural, leisure and entertainments experiences in destination travel. Lastly, Cluster 5, which is called Relaxed Leisure Tourists, is in the middle of Juice Bars and Religious Institution and lower in Restaurants and Theatres, thus indicates those tourists who enjoy less vibrant leisure experiences like quieter places though still some kind of activities and socialization.

All in all, these clusters can adequately represent the diversity of the tastes of tourists in the data. The findings point out that besides the nature of attractions that the tourists visit, they can be categorized based on their levels of engagement and degree of activity. Such an in-depth segmentation can guide tourism planners, marketers and service providers to customize experiences and resources to meet individual needs and interests of the various types of tourists and eventually increase visitor satisfaction and maximize destination management.

B. Hierarchical Clustering

The standardized tourist data was subjected to hierarchical clustering where 6 clusters were used to calculate the clusters using the dendrogram and the maximum cluster criterion. The score of 0.118 in the silhouette also means moderate separation of clusters meaning that although the method elicits some structure to data, there is an observable overlap between groups. This score is a little bit lower than silhouette score of 0.162 of KMeans, indicating that hierarchical clustering yields meaningful clusters, albeit with less clear-cut boundaries, that is typical of high-dimensional or continuous data sets with smooth variations in tourist behaviours.

The cluster centroid analysis with descriptive labels can further give information regarding the nature of individual tourist groups. Cluster 1 titled Relaxation and Culture-Oriented Tourists have high agreeableness in outdoor and natural facilities e.g. Parks and Beaches and moderate activity in cultural facilities e.g. Museums and Art Galleries. Such tourists seem to be more inclined towards a low intensity, relaxing leisure experience which is accompanied by cultural touring. Cluster 2, Social and Nightlife Seekers, is characterized by high scores in Juice Bars, Dance Clubs, and Resorts, moderate activity in Restaurants, indicating that the customers are young adults who are interested in nightlife, socialization and nightlife orientated entertainment. Cluster 3, Spiritual and Nature-Centred Tourists is differentiated by high scores on the religious Institutions and Parks, moderate scores on Art Galleries and Beaches, which mean that visitors focus on spirituality and nature-oriented tourism.

Cluster 4, Well-Rounded Experience Seekers demonstrates a moderate value on almost every feature implying tourists who do not have any one overwhelming preference and seek the combination of cultural, leisure and outdoor experiences. Cluster 5, "Pilgrimage & Nature Travelers," is more focused on religious destinations and outdoor activities and less on entertainment or dining, indicating that they are visitors who prefer spiritual or natural sightseeing. Lastly, Cluster 6, "Casual Leisure and Dining Tourists," is moderate in Juice Bars and Restaurants and low in nightlife and cultural attractions, as the users prefer light leisure and dining experiences with no need to engage in other activities intensely.

By and large, hierarchical clustering can be said to capture a wide range of tourist behaviours with ease categorizing visitors depending on their preferences and intensity of activities as well as their involvement in the same. The clusters emphasize the shared tourist trends, including social and entertainment tourists, and less popular ones, including nature-oriented and spiritually oriented tourists. This knowledge can be useful to tourism planners, marketers, as well as service providers so that they can offer targeted products and services, allocate resources more effectively and provide personalized experiences to satisfy the unique interests of each type of tourist. Although the clusters are a bit less well-separated than the clusters received through KMeans, hierarchical clustering offers a complementary point of view concerning the organization and the heterogeneity of tourist preferences.

7. Limitations

Although the clustering analysis provided powerful information on the tourist behaviour, it has a number of limitations that one ought to bear in mind when interpreting the findings.

A. Scaling and feature selection sensitivity:

Both hierarchical clustering and K-Means are based on distance information, and, thus, they are very sensitive to the size of the input features. Even though Min-Max scaling was used to equalize the contribution of features, there would be different ways to scale the features (e.g., z-score normalization) or weight some features, which would change clusters assignments. Also, analysis was performed only on the available numerical features, which might not reflect all the relevant variables of tourist preferences, including season studies, personal motives, or socio-economic factors.

B. Selection of the number of clusters (k):

The clusters were determined to be six that were established through the elbow test of K-Means and that were used in hierarchical clustering so that such levels could be compared. Nevertheless, hierarchical clustering implies that it might be reasonable to have fewer or more clusters depending on the amount of granularity required. The selected k balances are interpretable and methodologically consistent but might not be indicative of all substructures in the data which are natural.

C. Cluster overlap and heterogeneity:

The hierarchical clustering resulted in overlapping and heterogeneous clusters as the hierarchical clustering showed a lower value of silhouette score. This implies that not all observations are evidently members of a particular cluster and this could decrease the accuracy of the segmentation when using the findings on specific strategies or recommendations. Even more compact clusters such as K-Means clusters have the restriction that the clusters assume sphericity and have an equal number of clusters.

D. Limitations of data and generalizability:

The research is conducted on the basis of available data only, and this data might not be the reflection of the bigger tourist groups. The results of the clustering might not be applicable to other contexts due to the influence of unobserved variables, data collection bias, or changes in tourist behaviour over time.

E. Algorithmic limitations:

The two clustering algorithms are subject to limitations. K-Means assumes convex clusters and can have problems with irregular clusters, whereas the hierarchical clustering is sensitive to outliers and does not make use of a global objective function. The factors have the capability of affecting the quality and interpretation of clusters.

Nevertheless, these constraints do not imply that K-Means and hierarchical clustering should not be used together to offer mutually supportive information. K-Means determines compact and well-separated segments, whereas hierarchical clustering brings out higher-level structural patterns, which provide a more detailed insight into the preferences of tourists as opposed to either approach.

8. Conclusion

The tourist behaviour clustering analysis gave intricate and broad results of the varied preferences of visitors. K-Means and hierarchical clustering were used on the scaled data using six clusters each to reduce methodological inconsistency and comparability. K-Means formed small, isolated groups, as it optimized within-group variability, while hierarchical clustering showed larger and more nested groups and more heterogeneous groupings, which K-Means could not explicitly discover among the tourists.

Cluster profile analysis identified significant patterns of behaviour. Some of the clusters included strong inclination towards nature-oriented activities like beaches, parks, and resorts; some of the clusters were more inclined on the cultural and entertainment attractions like museums, art galleries, theatres, and dance clubs. Other clusters showed more inclinations to religious or heritage-oriented activities, others were more balanced in various activities which confirms more generalized tourism behaviour. Smaller clusters had niche segments with unique preferences whereas, large clusters had common patterns.

The two approaches to clustering gave the view that was complementary. K-Means provided more accurate segmentation and more distinct boundaries, which was especially helpful in targeted marketing, personalized suggestions, or the definition of homogenous groups of tourists. Overlapping clusters and hierarchical clustering, although with a lower silhouette score, provide the better understanding of the hierarchical relationship among tourist preferences, macro-level groupings and structural patterns which can inform broader strategic decisions.

Some methodological considerations were also pointed out by the analysis. The formation of clusters is a very delicate attribute to the scaling of features, the choice of k, and the type of distance measures. Although the elbow method helped identify six clusters in the K-Means and was similar to the hierarchical clustering in terms of comparability, other options of k might uncover more substructures. In addition, the dataset in itself can be a poor reflection of all the factors impacting tourist behaviour, including temporal changes, budgetary constraints, or unmeasured individual motivations and this reduces the applicability of the results.

Nevertheless, by its limitations, the study shows the success of applying more than one type of clustering methods to investigate complicated behavioural datasets. Using a combination of K-Means and hierarchical clustering, the analysis strikes the right balance between the deep segmentation and the overview of the larger structural patterns, giving a global picture of the tastes of the tourists. These findings have practical implications on tourism management, marketing strategies and service design and allow the players to provide offerings to various segments of visitors as well as identify general trends in visitor behaviour.

To summarize, the paper has highlighted the importance of exploratory clustering to discover fine-grained and structural patterns in behavioural data, it demonstrates the potential to further refine the depth, interpretability, and applicability of findings gained out of complex datasets through the use of complementary clustering techniques. The given analysis can be followed by further work, including more behavioural or demographic and time-related variables, to make segmentation more accurate and enhance the comprehension of the tourist preferences better.

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